

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd November 2024

SP6 - Financial Derivatives

**Time allowed: 3 Hours 15 Minutes (09.30 – 12.45
Hours)**

Total Marks: 100

- Q. 1)** Derive the minimum and maximum value of a European Call and Put option. Also derive the minimum and maximum value of an American call and option. For which type of options, European or American, will the pricing bounds differ, and why?

[5 Marks]

- Q. 2)** Consider the following homogeneous reference portfolio in a synthetic CDS:

- The number of reference entities = 100
- CDS spread, $s = 150$ basis points
- Recovery rate, $f = 50\%$
- On a single name, the annual default probability is constant over five years and obeys the relation, $s = (1 - f) \cdot PD$, where PD is the probability of default.

Assuming that the defaults are all independent of each other, what is the expected number of defaulting entities over the next five years? Tranches up to what point would be entirely wiped out (i.e. lose 100% of the principal invested) by the expected number of defaulting entities?

[5 Marks]

- Q. 3)** A farmer has won a large contract to supply barley to a nationally operating distillery in 12 months' time. The farmer or his employees has no experience in derivatives markets. However, the farmer has heard of derivatives, in particular futures, that allow to reduce risks. He is contemplating the use of futures market to hedge against the risk of a drop in the market price from the current levels (this is considered to be high) in 12 months. The current market price of barley is Rs. 7.50 per kilogram.

- i. Given this background, how suitable is it for the farmer to hedge the risk using futures contracts. **(5)**
- ii. Assume that the contract referred to above is to supply 4,000 kilograms of barley at the market price at the time of supply. The present value at the current date of the production cost to supply this amount of barley is Rs. 25,000 and can be assumed as a constant. The continuously compounded risk-free interest rate is 4.0% per annum. If the farmer decides to be unhedged calculate the market price of barley in 12 months, at which the farmer will neither make any profit nor make any loss. Please state your assumptions clearly. **(2)**
- iii. The farmer's investment advisor suggests an alternate hedging strategy using options in the following way:
 - Buy 4,000 vanilla European barley put options with a strike price of Rs. 6.00 and a cost of 2 paise per option.
 - Short 4,000 vanilla European barley call options with a strike price of Rs. 8.00 and a cost of 37 paise per option.

- a. To implement the strategy above, how much will the farmer have to pay initially to set up this hedge. Please clearly state any assumptions you make. **(3)**
- b. Calculate, showing all workings, the market price of barley in 12 months' time, at which the farmer makes neither a profit nor loss if the farmer hedges the risk using options as above. State any assumptions made. **(5)**
- c. Explain, with reasons, how appropriate this option based hedge strategy is compared to going unhedged. **(2)**

[17 Marks]

Q. 4) Let the spot price of stock W be \$50. What is the value of a European Call option on the stock that has a Strike Price of \$50 and time to Maturity of 65 days. The implied volatility of the stock can be taken as 25% per annum and the risk-free interest rate can be taken as 6% per annum continuously compounded.

- i. Calculate the value of the European Call option using Black Scholes formula. **(1)**
- ii. Calculate the Delta and Gamma for the above-mentioned option. **(4)**
- iii. The writer has sold 1000 calls. To create a delta neutral portfolio:
 - a. How many shares should the writer buy? **(2)**
 - b. How much additional money does the writer need to invest / borrow? **(2)**
- iv. If the spot market price of W moves to \$52 the next day with the other terms remaining unchanged, how can the call writer ensure delta neutrality for the 1000 calls that has been written? **(3)**
- v. On the day the option is written, the writer wants to be both Delta and Gamma neutral for the 1000 calls he has written. To do this, he finds that another option (option 2) on the same underlying is available in the market at a price of 1.1466 and the delta and gamma for option 2 are 0.2965 and 0.0529 respectively. In this case,
 - a. How many new options (option 2) should the writer buy? **(2)**
 - b. How many shares should the writer buy? **(2)**
 - c. How much additional money does the writer need to invest / borrow? **(2)**
 - d. Comment on why the cash requirement is lower in this case as compared to the delta neutral case. **(2)**

[20 Marks]

- Q. 5)** XYZ Insurance Company Ltd. sells single premium products such that returns on certain block of assets (called Growth fund) are used to increase the policyholder payouts (policyholder share); i.e. if growth fund experiences a total return of 10% then the policyholder payouts (hence companies' liability) will also grow by 10%. The rest of the company's money is in match fund which is invested in either government bonds or cash. There are also guaranteed payouts of premium at maturity, which if more than the policyholders' share at maturity will lead to a guarantee cost.

Balance sheet at 31/03/2024 for the company is given below:

Assets	
Growth Fund	1000
Match Fund	600
Total Assets	1600
Liabilities	
Policyholders share	1200
Guarantee Costs	300
Total Liability	1500
Free Capital	100

Asset allocation of the growth fund is

Initial Asset Mix	
Equity	30%
Corporate Bond	30%
Government Bond	30%
Property	10%

- i. You want to price the guarantee using Monte-Carlo simulation. Which are the risk drivers that you would recommend for the Monte – Carlo simulation? Suggest some models and inputs required to model the above recommended risk drivers. Further, explain how you will allow for interaction between them. **(4)**
- ii. Construct a stressed balance sheet for 50% fall in equity market & 50% rise in equity market. Delta for cost of guarantee with respect to change in policyholder share can be taken as -12% (minus 12%). State for which of these scenarios the free capital would be lower and why. **(6)**
- iii. A Fund manager expects property value to fall in coming months. Also, the company has physical property holding that is not possible to be sold in such a short time. As a solution, one of your peers has suggested creating a notional property exposure of 100 in growth fund and a matching notional negative property in match fund. Scenario test (20% property up and 20% property down) the suggested solution and argue whether proposed solution is appropriate if we expect fall in property. State what could be possible risk involved. **(6)**

[16 Marks]

Q. 6) ABC Bank's treasury uses the Black model to value vanilla interest rate options (caps, floors and swaptions). ABC bank is looking to extend its range of products into more exotic interest-rate swaps and options denominated in Euro, and is considering building a term structure yield curve model for pricing and hedging these instruments.

- i. Outline the reasons for choosing a full yield curve model in this situation. **(2)**
- ii. What are the desirable features of the chosen model? **(4)**
- iii. Explain the particular relevance of a "no arbitrage" condition in this context. **(2)**

The Hull-White one-factor yield curve model projects the short-term rate r according to the formula:

$$dr = a(t)(b(t) - r)dt + \sigma(t)dz$$

where $\sigma(t)$ is the time-dependent short-rate volatility and dz is a standard Brownian motion. The bank has decided to use an implementation that sets parameters $a(t)$ and $\sigma(t)$ as constants a and σ respectively for all t .

- iv. Describe how you would construct a trinomial rate tree for this model, calibrated to the term structure of zero-coupon bond prices and caplet volatilities. You do not need to derive the branching equations. **(7)**
- v. Discuss the extent to which your Hull-White implementation will fulfil the desirable criteria you identified in (ii) to help ABC Bank. **(5)**

[20 Marks]

Q. 7) You are an investment actuary. Your Chief Investment officer (CIO) who is an experienced investment professional (but new to derivatives) is thinking of buying a one year at the money call option on the NIFTY. He has asked you to calculate the fair value of the call option as on 30th Sep 2024, before seeking quotes on the option from the market.

You have decided to use the Black Scholes option pricing model for calculating the price of a NIFTY one-year call, using the following data and taking risk free rate as 6% per annum.

As on 30 th Sep	NIFTY at end
2011	5,826
2012	5,296
2013	5,683
2014	6,704
2015	8,491
2016	7,738
2017	9,173
2018	10,114
2019	11,624
2020	8,598

As on 30 th Sep	NIFTY at end
2021	14,691
2022	17,465
2023	17,360
2024	22,327

- i. You have reported your calculations to the CIO. The CIO has asked you whether your pricing model considers the long term drift of the NIFTY in the calculation. What will be your response? **(4)**
- ii. Seeing your calculated option price, the CIO remarks that the calculated price looks low and it is unlikely that someone will be willing to offer the option at the calculated price. Critique his comment. **(7)**
- iii. How do market participants who would be writing this option tackling the issues highlighted in part ii. **(6)**

[17 Marks]
